DRAFT

Discretizing Linear Systems

9 Discretizing Solutions of ODE's

I) Euler Methods and Error Analysis

A) Introduction

Euler's Method

$$y_{n+1} = y_n + h f(x_n, y_n)$$

B) A comparison

Given $y' = 2xy$ $y(1) = 1$ we can get an analytical solution of

$$y = e^{x^2-1}$$

We could also employ Euler's Method to get an approximate solution!

C) Improved Euler's Method

$$y_{n+1} = y_n + h \frac{f(x_n, y_n) + f(x_{n+1}, y^*_{n+1})}{2}$$

Where

$$y^*_{n+1} = y_n + h f(x_n, y_n)$$

II) Runge-Kutta Method

A) Introduction

One of the most well known and accurate methods is the fourth order Runge-Kutta method.

B) Fourth Order Runge-Kutta (RK4)

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \tfrac{1}{2}h, y_n + \tfrac{1}{2}hk_1)$$

$$k_3 = f(x_n + \tfrac{1}{2}h, y_n + \tfrac{1}{2}hk_2)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

III) Exact Discretization

This section picks up on notes from LMC class.

Consider the system

$$\dot{x} = Ax \qquad x(t) = x_0 e^{A(t-t_0)}$$
$$x(t_0) = x_0$$

Note this is just for and LTI system!

For the controlled system we get

$$\dot{x} = Ax + Bu \longrightarrow x(t) = \underbrace{e^{A(t-t_0)} x_0}_{\text{Initial/ Transient Part}} + \underbrace{\int_{t_0}^{t} e^{A(t-\tau)} B(\tau) u(\tau) d\tau}_{\text{Controlled Part}}$$
$$x(t_0) = x_0$$

This is an exact, continuous solution. Can we discretize this without any loss in fidelity?

Thinking about a real control system, it is very likely that a control will be held for so time $t > 0$, i.e. held constant. Let us rewrite the previous equation taking into account this time where the control is not changed.

$$x(t+T) = e^{A(\underline{t}+T-\underline{t})} x(t) + \int_{t}^{t+T} e^{A(t+T-s)} B u(s) ds$$

$$x(t+T) = A^T x(t) + \int_t^{t+T} e^{A(T+t-s)}\, ds \; \underbrace{Bu(t)}$$

Why can we pull this out?

Lets pause and visit the question here. We said we are concerned on intervals where the control is constant. Because of that $Bu(t) = Bu(T)$. We can then pull it out of the integral because $Bu(t)$ is constant!

Now lets rewrite some things to simplify.

Let $\tau = s-t$ and $d\tau = ds$

$$x(t+T) = e^{At} x(t) + \int_t^{T+t} e^{(T-\tau)}\, d\tau \; Bu(t)$$

And because our system is LTI, we can drop the t's from the integral.

Let $\tau = s-t$ and $d\tau = ds$

$$x(t+T) = e^{At} x(t) + \int_0^{T} e^{(T-\tau)}\, d\tau \; Bu(t)$$